

LiPo BATTERY FIRE

Lessons Learned

This Lessons Learned document focuses on an incident in which a lithium polymer battery and its charger ignited during the charging process. No one was injured, but the battery and charger were destroyed. It demonstrates the fire hazards associated with such batteries and associated equipment.

What Happened?

A researcher was charging an 11.1 Volt Hildow Sport Power lithium polymer (LiPo) battery using a HobbyFans LiPro Balance charger. The lab had been using the battery and charger for 1-2 years without issue, but the researcher had only needed to charge the battery once before. At around 11 AM on the day of the incident, the researcher connected the battery to the charger using the charger's two leads. The researcher turned the charger on, and the charger displayed an error message indicating that it could not detect contact through one of the leads. The researcher restarted the charger; this time, no error message was displayed. The researcher checked that the charger was set properly (it could be set to charge several different types of battery), placed the battery and charger on their lab bench, and left the lab.

At around 4 PM the same day, several lab members were working in the lab (though the initial researcher was not present at that time). Normally once charging completed, a loud beeping noise would sound, but no researchers heard this between 11 AM and 4 PM. Suddenly, the researchers heard a loud pop and when they looked, they saw the battery on fire, with flames reaching around 2 feet in height. The Lab Safety Coordinator (LSC) was present at the time, and they ordered those present to evacuate. The LSC then went to the hallway to get a fire extinguisher. As they left the lab, the flames had been dying down, but while in the hall they heard another loud pop and when they returned the flames were again around 2 feet tall. The LSC discharged the CO2 extinguisher on the battery and the flames went out.

The LSC then called EH&S's front desk number. EH&S staff advised the lab to call 911; meanwhile, two EH&S staff members took CellBlock (a lithium/LiPo battery extinguishing material) and a metal can for debris to the lab. Palo Alto Fire Department arrived and assessed that the fire was fully extinguished. The EH&S staff then placed the battery and charger in the metal can, covered them with CellBlock, and transported the items to EH&S.

Why Did This Happen?

Due to the damage caused by the fire, it is not possible to determine with certainty the cause of the incident. Based on the errors encountered earlier in the day, it is possible that the charger could not detect when charging was complete and overcharged the battery, causing it to heat up and ignite.

It is challenging to determine if a given model of battery charger is safe. There is currently no Nationally Recognized Testing Laboratory (NRTL) with a standard for certification of small device lithium ion battery chargers (Underwriters Laboratories has a standard that applies only to chargers for electric vehicles and

industrial-scale batteries). This means that in practice, the only way to judge the quality and safety of a charger before purchase is to read product reviews, hoping that any safety issues have been reported. The researcher using the charger was not the person who purchased it (that person had since left the lab), so it is not known how thoroughly the purchaser researched the device or if reviews reporting safety issues had been posted at the time of purchase. The batteries themselves were marked as UL listed.

Due to this unpredictability, LiPo batteries should be monitored and protected while charging. While the battery was not left unattended, it was not being actively monitored, which meant that the lab did not notice that it was overheating. During testing of a replacement charger, the researcher closely monitored the charging process and noticed the charger was beginning to overheat and the battery was slightly overcharged, stopping the process before it could worsen. The researcher later purchased a more sophisticated charger with under-voltage sensing and auto-timeout, which has worked well since then.

Another way to mitigate the unpredictability of charger quality is to place charging batteries in specially designed pouches, bags, or boxes meant to insulate and contain charging lithium batteries. The lab uses these bags for long-term battery storage, but the battery was not placed in one while it was charging during this incident. While such devices may not prevent a battery from overheating, they can prevent the fire from spreading beyond the battery.

What Went Right?

Ultimately, no one was injured and the incident only resulted in minor property damage. A number of factors contributed to ensuring this incident was not more serious:

- The researcher's workspace where they placed the charging battery was clean and organized, free of combustible materials. Had there been significant amounts of clutter around the battery and charger, the fire most likely would have spread and grown.
- The researcher made a good decision to charge the battery during the day while the lab was occupied. While the fire did not spread, if it had, researchers would have called 911 right away and the fire's spread would have been mitigated.
- The lab members promptly evacuated when the fire started.
- The LSC successfully extinguished the fire.
- The incident was promptly reported to EH&S.

How to Prevent and Respond to Accidents Like This

EH&S has a number of recommendations for how labs can prevent incidents like this:

- Researchers should purchase high-quality equipment for their work.
 - When possible, researchers should purchase NRTL-certified equipment models.
 - When NRTL certification is not available for a class of equipment, researchers should purchase equipment directly from reputable manufacturers when possible, or from reputable vendors (e.g., Grainger, McMaster-Carr).
 - Researchers should read available product reviews, looking for any reports of hazardous malfunctions.
 - When purchasing lithium ion batteries, purchase a charger from the same manufacturer that

makes the battery, i.e., do not purchase 3rd-party chargers unless necessary.

- When purchasing chargers, researchers should look for ones with features such as cell balancing, under-voltage sensing, and timer shutoffs.
- For an extra layer of security, researchers may purchase a timed-shutoff outlet to power the charger if the charger itself does not have this feature.
- Take precautions when charging lithium ion batteries.
 - Charge lithium ion batteries inside protective pouches, bags, or boxes designed for the purpose of charging such batteries safely.
 - Charge lithium ion batteries at a low current.
 - Monitor the battery during charging, checking regularly to ensure it is not overheating. Unplug the charger immediately if the battery is overheating. Never leave a battery charging unattended.
 - Make sure the area around the battery is clean and free of any unnecessary combustible materials.
 - Do not charge lithium ion batteries near room exits, as a fire could then block a path of egress.
 - Do not over-discharge lithium ion batteries. Make sure there are sensor circuits in the device using the battery which can shut off operations if the battery reaches a voltage that it cannot be safely charged back up from.
- Be prepared for lithium ion battery fires. Ideally you should have a metal bin and a fire suppression material designed for use with lithium ion batteries readily available, though an ABC or CO2 fire extinguisher may be effective for small fires.
 - When in doubt, evacuate the area, close the door, pull the nearest fire alarm, and call 911 once outside. Never try to fight a fire if you do not feel comfortable doing so.
 - Make sure that there is a fresh fire extinguisher within direct access of the lab space and that lab members comfortable using it are aware of its location.
- If you need EH&S's help during an incident, call our 24-hour hotline at 650-725-9999. Have lab members add this number to their phone's contacts, as search engine results for EH&S's phone number often return EH&S's front desk number, which is only monitored during normal business hours.